MIL-C-18636A(OS)
13 February 1984
Superseding
MIL-C-18636(NOrd)
16 May 1955

MILITARY SPECIFICATION

CAPS, PROTECTING, WATERPROOF
(Projectile Fuze Covers)

This specification is approved for use by Naval Sea Systems Command (OS) Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers waterproof protecting caps for various Marks and Mods of 5-inch, 6-inch, and 8-inch through 16-inch projectiles.

1.2 Classification. The waterproof protecting caps covered by this specification shall be of the drawn cup, attached lug type, and of the following classes, as specified (see 6.2).

Class 1 - Mark 3 Mod 0 steel cap (Drawing No. 329747)
Class 2 - Mark 4 Mod 0 steel cap (Drawing No. 437782)
Class 3 - Mark 4 Mod 1 brass cap (Drawing No. 583511)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 Ammunition, General Specification for

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station (524), Indian Head, MD 20640, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this documents or by letter.

FSC 8140

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedure and Tables for Inspection by Attributes
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-1169	Packaging, Packing and Marking for Shipment of Inert Ammunition Components
DOD-STD-2101	Classification of Characteristics

DRAWINGS

Naval Sea Systems Command (Naval Ordnance Systems Command) (Code Ident 10001)

329747	Waterproof Protecting Cap, Mk 3 Mod 0
437782	Waterproof Protecting Cap, Mk 4 Mod 0
583511	Waterproof Protecting Cap, Mk 4 Mod 1

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Non-government documents. The following documents form a part of this specification. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

B154 Mercurous Nitrate Test for Copper and Copper Alloys

B633 Electrodeposited Coatings of Zinc on Iron and Steel

(Copies of ASTM standards are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 18120).

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3. REQUIREMENTS

3.1 General. The dimensions, finishes, tolerances and other design characteristics of the waterproof protecting caps shall conform to the applicable drawing(s) for the class of cap required (see 6.2). In addition the requirements of MIL-A-2550 shall apply as applicable.

3.2 Material. The caps shall be manufactured from steel or brass as specified on the applicable drawing for the respective class of cap required (see 6.2).

3.3 Cups. The cups shall be drawn and machined in accordance with the best manufacturing practice and shall be free from wrinkles, cracks, holes or any other defect that will tend to make the cup unsuitable for the purpose intended (see 6.1). The brass cups shall be free from susceptibility to season cracking.

3.4 Lugs. The lugs shall be manufactured from material specified for the class of cap required (see 6.2). The surface of the lug intended for contact with the cup shall conform to the contour of the outside surface of the cup at point of contact and shall be free from cracks, pits or otherwise weakened cross-sections that will tend to cause failure when subjected to the torque test of 4.4.2 or the axial load test of 4.4.3.

3.5 Torque test. After being attached to the cup each individual lug shall be capable of withstanding torque specified on the drawing. No evidence of breakage, permanent distortion or loosening shall occur when any cap is subjected to the test of 4.4.2.

3.6 Axial load test. Each individual lug shall be capable of withstanding axial load specified on the drawing. No evidence of breaking, permanent distortion or loosening shall occur when the cap is subjected to the test of 4.4.3.

3.7 Stress relief.

3.7.1 Class 1 and Class 2 caps. Class 1 and Class 2 caps shall be stress relieved at 1100 to 1200 degrees Fahrehheit after assembly to minimize warpage and objectionable stresses in the finished caps.

3.7.2 Class 3 caps. Class 3 caps shall be annealed after assembly, and prior to final machining, to insure freedom from season cracking, and shall successfully pass the stress test of 4.4.1.

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3.8 Plating. Class 1 and Class 2 caps shall be zinc plated in accordance with ASTM B633, Class FE/ZN 12, Type II. Following the plating operation, caps shall be subjected to a supplementary chromate treatment so as to produce a continuous, smooth protective film distinctly colored either iridescent bronze, iridescent yellow, olive drab, or brown.

3.9 Cleaning. All oil, grease, drawing and cutting compounds, dirt, excessive scale, solder and other foreign substances shall be removed from all surfaces of the caps. Brass caps may be pickled and rinsed.

3.10 Identification marking. Caps furnished under this specification shall be legibly and permanently marked in accordance with the requirements of MIL-STD-130, and the applicable drawing.

3.11 Workmanship. Workmanship shall be such as to produce a uniform and acceptable product in accordance with the detailed requirements of the applicable drawing and specifications.

3.12 First article. A first article sample shall be inspected in accordance with Section 4.4 prior to regular production in order to demonstrate the adequacy of the contractor's processes and procedures in achieving the performance that is inherent in the design. All parts and materials including packaging shall be the same as will be used for regular production and shall be obtained from the same source of supply. A first article sample or portion thereof, as directed by the procuring agency, shall be resubmitted whenever any of the following conditions occur:

a. There is a lapse in production for a period in excess of 90 days.

b. A change occurs in manufacturing process, materials used, drawings, specifications, or source of supply such as to significantly affect product unformity as determined by the responsible technical agency.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. The provisions of MIL-A-2550 shall also apply.

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4.2 Classification of characteristics. Characteristics to be verified are classified as Minor in accordance with DOD-STD-2101.

4.3 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection

b. Quality Conformance Inspection

4.3.1 First article inspection.

4.3.1.1 First article sample submission. A sample consisting of 30 completed caps shall be inspected and provisionally accepted at the source. Specimens shall then be forwarded to a Government laboratory or other facility specified in the contract to be inspected for all drawing and specification requirements in accordance with Table I.

4.3.1.2 First article acceptance. If any specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate its inspection upon any failure of any specimen in the sample to comply with any of the stated requirements.

TABLE 1. CHARACTERISTICS/TESTS CORRELATION

Requirement	Characteristic	Test Method	Test Sampling	
			First Article	Quality Conformance
3.1	Dimensional	Gage	100%	AQL 2.5% per MIL-STD-105
3.5	Torque test	4.4.2	100%	25 per lot selected at random
3.6	Axial load test	4.4.3	100%	25 per lot seleted at random
3.7.2	Stress test (Destructive)	4.4.1	100%	2 per lot
3.11	Workmanship	Visual	100%	AQL 2.5 per MIL-STD-105
5	Packing and Marking	Visual	100%	AQL 2.5 per MIL-STD-105

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4.3.2 Quality conformance inspection. The quality conformance inspection shall verify that all characteristics of Section 3, REQUIREMENTS, and Section 5, PREPARATION FOR DELIVERY, of this specification have been met. Table I correlates characteristics and corresponding tests. Quality conformance lots shall consist of 5000, minimum, completed caps including those caps required for tests. At the discretion of the Government, the lot and test sample size may be increased after the first five lots.

4.4 Acceptance tests.

4.4.1 Stress test. (Class 3 caps only. Destructive test). The completed brass caps shall be thoroughly cleaned with a suitable solvent and then immersed in an aqueous solution containing 1 percent by weight of mercurous nitrate and 1 percent by weight of nitric acid (specific gravity 1.42) for 15 minutes at 70 ±5 degrees fahrenheit. At the end of this period, all interior and exterior surfaces of the caps should have become completely covered with metallic mercury. Repeat operation if necessary to completely cover all surfaces with mercury. The caps shall be held in this condition for 24 hours and at the expiration of this time shall be free from cracks. Failure to pass this test shall be cause for rejection of the lot.

WARNING

Mercury is a definite health hazard. Proper safety precautions and disposal procedures shall be observed. Reference ASTM B154

4.4.2 Torque test. The completed caps shall be screwed onto a projectile or suitable fixture and the torque gradually applied to one lug in the direction of screwing the cap on further until the prescribed torque load per lug is reached. (see 6.3) The test shall then be repeated in the opposite direction but the cap shall not be allowed to unscrew. The procedure shall then be repeated for each of the remaining three lugs. Failure of any lug to meet the torque requirement at the conclusion of this test shall be cause for rejection of the lot.

4.4.3 Axial load test. After successful completion of the torque test, the same specimens shall be constrained in a suitable fixture for the application of an eccentric axial load to each upturned tang of the lugs in the direction illustrated in Figure 1. The specified load shall be gradually applied to one lug at a time. The load may be equally distributed over the surface of the tang. Failure of any lug to meet the load requirement shall be cause for rejection of the lot.

4.4.4 Disposition of test samples. The sample caps which have been submitted to mercurous nitrate stress test (see 4.4.1) shall be scrapped. The caps which satisfactorily pass the torque test of 4.4.2 and loading test of 4.4.3 may be accepted as part of the production lot, provided they have not been damaged or distorted in the test and still meet all dimensional requirements. However,

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if a lot fails to pass the stress or torque test and a retest is granted by the Government, none of the sample caps originally used in these tests may be used in the retest sample.

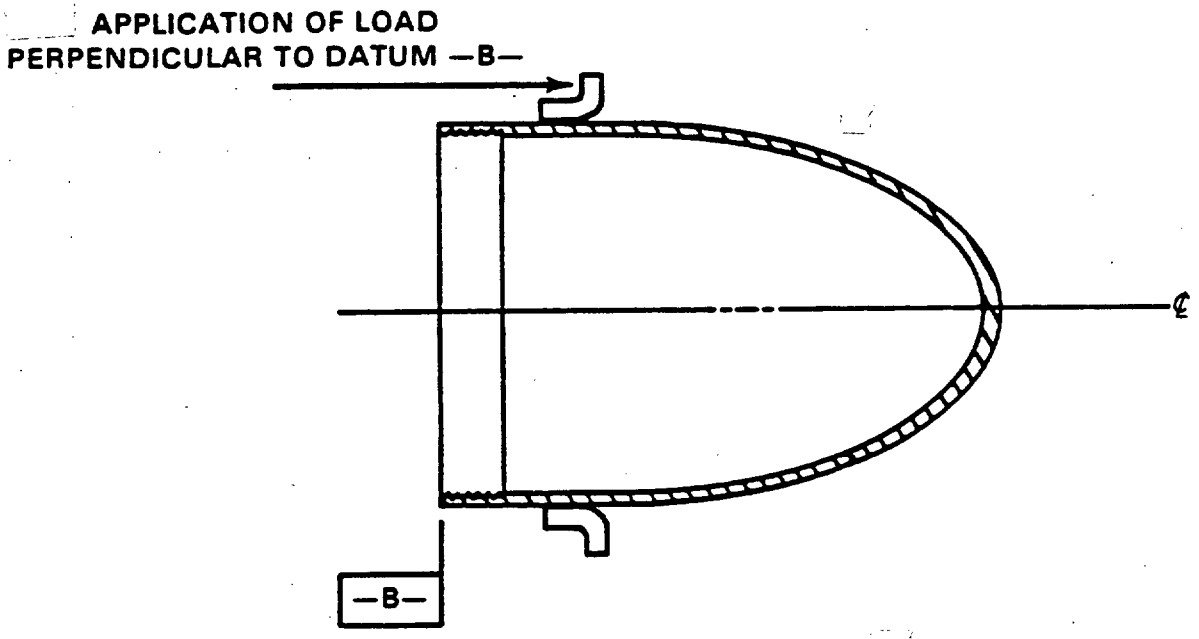


Figure 1. Axial Load Tests for Class 1 and Class 2 Caps

5. PREPARATION FOR DELIVERY

5.1 Packaging. Not applicable.

5.2 Packing. Unless otherwise specified by the contract, order or requisition, waterproof protecting caps shall be packed for domestic shipment in accordance with MIL-STD-1169, Level B. Caps shall be completely dry prior to packing and shipment.

5.3 Marking. In addition to any special marking required by the contract, order or requisition (see 6.2), shipping containers shall be marked in accordance with MIL-STD-1169, Level B.

6. NOTES

6.1 Intended use. When properly assembled with applicable projectiles, the waterproof protecting caps covered by this specification are intended to project the fuze and nose of these projectiles from damage by water and other causes.

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6.2 Ordering data. Procurement documents for waterproof protecting caps should specify the following:

a. Title, date and number of this specification

b. Class of cap required (see 1.2)

c. The contract, order or requisition should also contain pertinent information on Inspection Gages.

d. Special packing and shipping instructions, other than specified in this specification.

e. Special markings when required.

6.3 Torque and axial load tests. If load per lug is not indicated on the drawing, it may be calculated by dividing the total load by the number of lugs.

6.4 Changes.. Due to the extensiveness of this revision, the margins of this specification have not been marked with any symbols as is customary to indicate where changes from the previous issue were made.

6.5 Patent notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regared by implication or otherwise as in any manner licensing the holder or any person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

Preparing Activity
Navy - OS

(Project No. 8140-N628)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-C-18636A(OS)	CAPS, PROTECTING, WATERPROOF (Projectile Fuze Covers)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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